25

Version No.				
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ROLL NUMBER						

COMPUTER SCIENCE
HSSC–I
SECTION – A (Marks 13)

Time allowed: 20 Minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed.

Do not use lead pencil.

حصہ اول لازمی ہے۔اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔لیڈ پنسل کا استعمال ممنوع ہے۔

Answer Sheet No. __________

ہر سوال کے سامنے دیے گئے، کر یکولم کے مطابق درست دائرہ کو پر کریں۔ **Invigilator Sign. __________**

Fill the relevant bubble against each question according to curriculum: **Candidate Sign. __________**

	Question	A	B	C	D	A	B	C	D
1.	Which of the following logic gates produces a High (1) output only when all of its inputs are High (1)?	AND gate	OR gate	NOT gate	XOR gate	○	○	○	○
2.	Which computational process helps in solving a large problem by breaking it down into smaller parts?	Pattern Recognition	Decomposition	Abstraction	Recursion	○	○	○	○
3.	A researcher collects students' exam scores and wants to visually represent the frequency distribution of marks. Which graph is the most suitable?	Pie chart	Line graph	Histogram	Scatter plot	○	○	○	○
4.	Which statement correctly explains the purpose of the Software Development Life Cycle (SDLC)?	To design the physical hardware of computers	To manage the flow of data in a network	To organize and manage the stages of software development	To secure communication using encryption	○	○	○	○
5.	Which is an example of a data-collection approach for primary research?	Reading old newspapers	Conducting surveys	Copying from online sources	Watching movies	○	○	○	○
6.	Which of the following is the correct symbol used for a single-line comment in Python?	//	/*	>>	#	○	○	○	○
7.	In binary search algorithm, what happens if the middle element is greater than the target value?	The search continues to the right half of the list	The list is rearranged	The search continues in the left half of the list	The search stops immediately	○	○	○	○
8.	Which of the following explains the main purpose of experimental design in data science?	To collect as much random data as possible	To increase the size of the dataset	To plan and structure data collection and testing for valid conclusions	To store large datasets in memory for future investigation	○	○	○	○
9.	Which of the following is a technology that enables Blockchain to function securely?	Cloud Storage	Image Processing	Web Browsers	Cryptography	○	○	○	○
10.	Why should entrepreneurs iterate a prototype after testing?	To refine the solution	To impress investors only	To make the prototype more colorful	To avoid customer involvement	○	○	○	○
11.	Which of the following is an example of a reliable information source?	Random social media post	Peer-reviewed journal article	Anonymous blog	Entertainment website	○	○	○	○

12.	The output of the following Python code is:
	`x = 10`
	`y = 3`
	`print(x % y)`
		30	7	1	0	○	○	○	○

13.	An entrepreneur designs a mobile app for food delivery and asks a group of students to use it for one week. What entrepreneurial process is being practiced here?	Brainstorming only	Prototype testing	Final product launch	Single seller market	○	○	○	○

——1HA-II 25007——

COMPUTER SCIENCE HSSC–I

Time allowed: 2:40 Hours **Total Marks Sections B and C: 62**

SECTION – B (Marks 42)

Q. 2 Answer the following parts briefly. **(14 x 3 = 42)**

(i)	Why is algorithm efficiency important? Explain briefly.	03	OR	How does a debugger help programmers? Justify.	03
(ii)	What is the use of statistical modelling? Give example.	03	OR	What is meant by the Internet of Things (IoT)? Also give one real-life application.	2+1
(iii)	Is design phase an important part of the SDLC? Justify your answer.	1+2	OR	Briefly explain the difference between parameter and statistics with an example.	1.5 + 1.5
(iv)	Compare low-fidelity and high-fidelity prototypes with examples.	1.5 + 1.5	OR	A university network is failing frequently due to high traffic. Suggest a network topology that improves reliability and justify your choice.	1+2
(v)	Why do programmers use libraries in Python instead of writing all code from scratch?	03	OR	A software company is developing an online banking system. Which SDLC model (Agile or Waterfall) is more suitable? Justify your answer.	1+2
(vi)	What is the impact of the digital divide? Give example from everyday life.	2+1	OR	Briefly explain public cloud. Give one example.	2+1
(vii)	How does bubble sort differ from insertion sort in the way they arrange elements?	1.5 + 1.5	OR	Why is it better to use functions rather than writing one long program? Provide three reasons.	1x3
(viii)	How is indentation important in Python? Elaborate with example.	03	OR	Differentiate between analog and digital signals. (any three)	1x3
(ix)	What is meant by a survey in digital data collections? Give one example of a survey question.	2+1	OR	What is a prototype in entrepreneurship? Give example.	2+1
(x)	Construct a truth table for the Boolean expression: **$F = A.B + \overline{A}.C$**	03	OR	Write Python statements considering turtle graphics to perform following actions: a. Change the pen colour to blue. b. Rotate the turtle 90 degrees to the right. c. Move the turtle forward by 50 pixels.	03
(xi)	Write down any three advantages of infographics.	1x3	OR	What is secondary source of information? Give example.	2+1
(xii)	Why do organizations need cybersecurity frameworks? Give any three reasons.	1x3	OR	Differentiate between supervised and unsupervised learning with examples.	1.5 + 1.5
(xiii)	Suppose a list = [2, 4, 6, 8, 10]. Write a Python program to find the sum of all elements in the list.	03	OR	Evaluate the following expression in Python using the proper order of operations: `4**2-6%3+8/4*2`	03
(xiv)	Differentiate between black-box testing and white-box testing in software development with example.	1.5 + 1.5	OR	How can data visualization help in interpreting large datasets? Give example.	1+2

SECTION – C (Marks 20)

Attempt the following questions. **(4 x 5 = 20)**

Q.3	Simplify the Boolean function F, using Kamaugh map. **$F=\overline{A}\,\overline{B}\,\overline{C}+\overline{A}B\overline{C}+A\overline{B}\,\overline{C}+A\overline{B}C+AB\overline{C}$** Also construct logic circuit for the simplified expression.	05	OR	Develop a Python program using functions that takes two numbers from the user, and prints their product, sum and difference.	05
Q.4	Discuss the concept of experimental design in data science. How it helps in differentiating between correlation and causation? Explain with the help of an example.	1+2 +2	OR	Compare Symmetric and Asymmetric encryption with respect to efficiency, security, key length, large data handling and real-life example.	1x5
Q.5	Apply the binary search algorithm step by step to find the number 42 in the sorted list: [12, 18, 25, 29, 35, 42, 47, 53, 60]	05	OR	Write a Python program that takes an integer as input and prints the multiplication table of that number up to 10 using for loop.	05
Q.6	What is Blockchain technology? Explain decentralization and transparency in blockchain with real-life examples.	1+4	OR	What is increased connectivity in computing? Discuss its environmental and cultural impact on society with examples.	1+4